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EMULGEL-BASED DRUG DELIVERY SYSTEMS FOR TOPICAL ANTIFUNGAL APPLICATIONS: A REVIEW

Rachael Amin Elia Lupe - Research Scholar | rachaellupe6@gmail.com

Department of Pharmacy, Kalinga University, Raipur, Chhattisgarh

Dipak Bhadre - Assistant Professor | dipak.bhadre@kalingauniversity.ac.in

Sandip Prasad Tiwari - Assistant Professor | sandip.tiwari@kalingauniversity.ac.in

Faculty of Pharmacy, Kalinga University, Raipur, Chhattisgarh

ABSTRACT:

Emulgel is an advanced topical drug delivery system designed to improve the delivery of antifungal drugs through the skin. It combines the advantages of emulsions and gels, enabling better solubility of hydrophobic drugs and enhanced skin penetration. Topical delivery avoids first-pass metabolism, improves patient compliance, and allows controlled drug release. The formulation depends on oil phase, aqueous phase, emulsifiers, polymers, and permeation enhancers. This review highlights formulation strategies, mechanisms of drug release, evaluation parameters, and recent advancements in antifungal emulgels, along with marketed products and future perspectives.

Keywords: *Emulgel, Topical drug delivery, Antifungal agents, Ketoconazole, Controlled release, Skin permeation, Nanoemulgel*

INTRODUCTION

Fungal skin infections are common in humid regions and can severely impact a patient's quality of life [1, 5]. Ketoconazole is a broad-spectrum antifungal that works by compromising fungal cell membranes; however, traditional delivery methods like creams often lack the penetration and stability needed for optimal results [1, 2]. Emulgels offer a modern solution by combining emulsions and gels to improve drug solubility and skin absorption. This dual system provides a non-greasy, stable, and highly effective way to deliver ketoconazole [3,4]. This review explores how emulgel formulations can overcome the barriers of traditional topical therapy to provide better clinical outcomes for superficial infections [1,3,5].

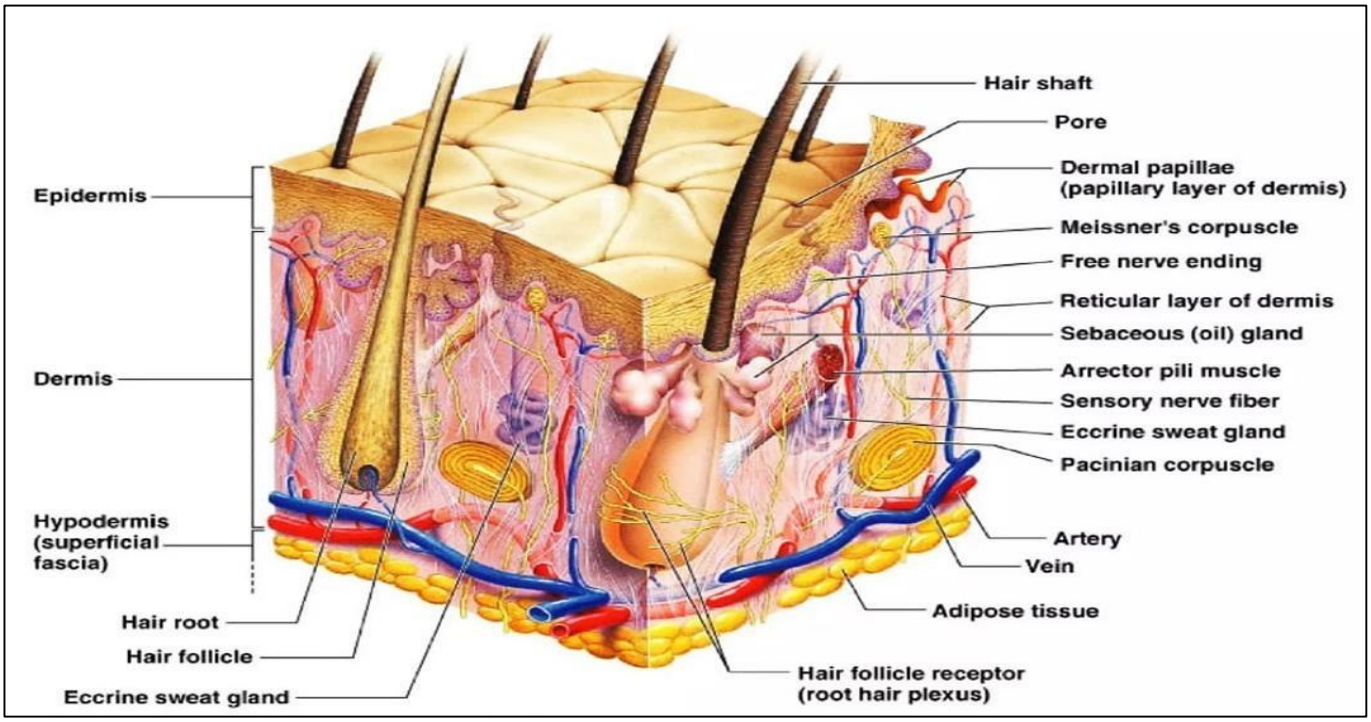


Figure 1: This figure shows the anatomy of the human skin

EMULGEL AS A TOPICAL DRUG DELIVERY SYSTEM

An emulgel is basically a newer take on topical drug delivery that pulls together qualities from both emulsions and gels [5,6]. The structure's more straightforward, you've obtain an emulsion, either oil-in-water (O/W) or water-in-oil (W/O), dispersed inside a gel matrix made from gelling agents like Carbopol, hydroxypropyl methylcellulose (HPMC), or xanthan gum. That combination creates a semi-solid you can use topically, and it works especially well for drugs that don't dissolve great in water [6,7,8]. The gel network does a few things at once: it keeps the emulsion droplets from breaking apart, makes the stuff easier to spread, and helps it stick around longer on your skin. Meanwhile, the emulsion part helps dissolve lipophilic drugs and lets them release

gradually. Emulgels aren't greasy, they go on easily, and honestly, patients like how they feel way more than traditional ointments or creams [7,8]. Because of how they're built and their properties, emulgels turn out to be really good for drugs like ketoconazole that hate water. They let the drug get deeper into skin, cause less irritation, and get better absorption where the infection actually is [8,9].

TOPICAL DRUG DELIVERY SYSTEMS

When you're dealing with localized skin infections, especially fungal ones, topical drug delivery systems are pretty much the better approach [17, 45]. They've gotten some real benefits, you get targeted delivery right where you need it, fewer systemic side effects, and honestly, patients tend to stick with them better. Drugs struggle to get through the stratum corneum, which is a real headache if you're working with hydrophobic compounds [5, 34].

Here is the thing about topical drug delivery: the whole point is getting a specific drug to work with and pass through the skin. The tricky part is actually getting it across that skin barrier [36,41]. There are basically two flavors of topical drugs, internal and external. Internal one work on mucous membranes, applied through the mouth, vagina, or rectal area. The external one get sprayed or rubbed onto your skin to hit the problem spot16, 15, 49].

Traditional gels like creams and ointments? They've got issues. They tend to feel greasy, they're not super stable, and the bioavailability isn't great, which means you don't get the therapeutic punch you're hoping for [44, 51].

CONCEPT AND EVOLUTION OF EMULGEL SYSTEMS

Emulgels came about because someone realized it, why not combine what works in emulsions and gels? Basically, you take an emulsion and work it into a gel base. You get this dual-controlled release thing going on [29, 37].

Over the past decade or so, people have really warmed up to emulgels because they actually work pretty well. The permeability gets better, they're more stable, not hard to make, and they look and feel nicer on skin. And then you've got these newer versions like nanoemulgels that push the whole drug delivery efficiency even further [8, 15,].

COMPONENTS OF EMULGEL FORMULATION

An emulgel is really a two-steps process you form from your emulsion, then fold it into a gel base. This allows maintenance of a constant drug level and promotes uniform distribution throughout [2, 13, 58]. Now people are doing much more crazy things using nanoemulsions to make drugs better absorbed. Usually either Carbopol 934 or 940 do the job as gelling agents [44, 56]. You select the oil phase depending on how well your drug truly dissolves in it. more frequently used compounds such as Caproyl 90, Labrafil and isopropyl myristate [1, 40]. You've got natural oils like mentha oil, clove oil and many of those have their benefits. To give it some semblance of stability, you add emulsifiers such as Tween 80 and Span 80. They shatter the surface tension and keep those oil micropuddles hanging out together just right [20,30]. The aqueous phase usually starts with purified water but if you are working on moisture sensitive drugs, propylene glycol or polyethylene glycol might be used in its place. This is really useful for preparations such as ketoconazole which does not solubilize well in water [42, 59].

ADVANTAGES:

a. Avoidance of first pass metabolism.

b. Avoidance of gastrointestinal incompatibility.

c. More site specific medium selective

d. Improve patient compliance.

e. Suitability for self-medication.

f. Utilization of drugs with short biological half-life and narrow therapeutic window

g. Ease of withdrawal from the drug as required;

h. Convenient, easy to apply process

i. Incorporation of hydrophobic drugs

j. Better loading capacity

k. Better stability

l. Feature of linen plant production & low preparation cost

m. Controlled release

n. No intensive sonication

DISADVANTAGES:

a. Skin irritations (contact dermatitis)

b. Potential for allergic reactions third with low permeability of few drug thru skin

d. Substances with larger particle size that is difficult to penetrate the skin

e. Bubble formation (bubbles forming when emulgel is made);

f. Stability issues (phase separation)

g. High surfactant concentration

h. Limited penetration for large molecules

i. High production costs for nanoemulgels

Antifungal therapy is needed for a wide range of ailments [22].

Hydrophobic (lipophilic) drugs such as Clotrimazole, Miconazole and Fluconazole deliverance of antifungal infectious diseases. Traditional gels are both visually soothing and quickly absorbed without the greasy finish, but have difficulty incorporating such water-insoluble agents. [23]

DUAL-RELEASE SYSTEM:

A emulgel offers a system of dual release in which the drug diffuses out from an outer semisolid material (emulsion) first and then serves as into three-dimensional gel network. This offers constant release and enhances the deposition of drug in target site.[35, 45]

BETTER PENETRATION:

Surfactants and an oily phase assist a drug to penetrate across the stratum corneum (skin's outer barrier) thereby increasing its bioavailability. Factors influencing drug absorption into the topical vehicle [8, 9]

PHYSIOLOGICAL FACTORS:

1. Skin thickness.
2. Lipid content.
3. The density of hair follicles.
4. The density of sweat glands.
5. Skin pH. 6. Blood flow.
6. Hydration of skin.
7. Inflammation of skin.

PHYSICOCHEMICAL FACTORS:

1. Partition coefficient.
2. The molecular weight (<400 Dalton).
3. The extent of ionisation (Only unionised drugs are absorbed well)
4. Effect of vehicles.

THE TWO BASIC COMPONENTS OF AN ANTI-FUNGAL EMULGEL:

To formulate a successful formulation, you will need to choose several active ingredients before all else:

- Aqueous Phase: The continuous phase will be purified in water or the alcohols.
- Oily Phase: Liquid Paraffin, Jojoba oil or Clove oil as oils to use in dissolving the hydrophobic antifungal drug.
- Gelling agents such as Carbopol 934,948 or HPMC are generally added to the formulation in giving viscosity and
- thixotropic (shear-thinning) properties. Emulsifiers (eg, Tween 80, Span 80) stabilize the combination to prevent
- segment separation.
- pH Adjusters: Triethanolamine or NaOH is used to neutralize the gel making sure that its ph level matches that of skin (approx.5. 5-6. 5).

PREPARATION METHODOLOGY

The formulation process usually consists of the following three main steps: [11, 19, 58]

1. Emulsion preparation The pure oil-soluble (drug, Span 80) and water soluble components (Tween 80, preservatives). were separately heated to 70–80°C and mixed in a continuous stirred tank for the resulting mixture O/W emulsion.
2. Gel Base: Gel base is prepared by dispersing the gelling polymer in water and neutralizing it to get consistency.

3. Incorporation: The emulsion previously formed is mixed with the gel base (generally in a 1 sampling anxiety ratio) using continuous stirring until a homogenous semi-solid emulgel is achieved.

EVALUATION AND QUALITY CONTROL:

Emulgels are a new type of topical formulation, they must go through an extensive testing process for both safety and efficacy:

- **Visual Inspection**: Color, homogeneous and absence of grit
- **Spreadability & Extrudability:** Indicates the ease with which a gel spreads over skin and how easy it is to extrude from tube.
- **β-Tests with Franz diffusion cells**: In-vitro Drug Release from adsorbent (using synthetic or biological membrane such as egg membranes for studying release kinetics).
- **Antifungal Assay:** Conducted by the ditch plate or agar well diffusion method against fungal strains such as Candida albicans.
- **Stability Studies:** The samples are stored at different temperatures (40°C/75% RH) for months, to observe the change in pH, viscosity and drug content.

ADVANTAGES OVER TRADITIONAL FORMULATIONS:

Feature	Traditional Ointment/Cream	Antifungal Emulgel
Texture	Greasy and sticky	Non-greasy and elegant
Stability	Prone to phase inversion	High stability due to gel network
Drug Release	Often slow/irregular	Controlled and sustained
Washability	Difficult to remove	Easily washable with water

The future of this field lies in Nanoemulgels, which use even smaller droplets (less than 100 nm) to further enhance skin penetration and therapeutic outcomes. [20]

RATIONALE FOR EMULGEL FORMULATION:

Emulgels that are formulated to address the drawbacks of traditional topical systems:

- Poor penetration of hydrophobic drugs
- Greasiness of ointments
- Instability of emulsions Advantages include:
- Controlled drug release
- High drug loading capacity
- Non-greasy and easily spreadable formulation
- Improved bioavailability

METHOD OF PREPARATION:

Emulgel is being prepared with three simple steps.

1. Methods of Preparation of gel base Carbopol + neutralizer
2. To prepare emulsion (oil + at 70–80°C aqueous phase)
3. Slow integration of emulsion in the gel while continuously stirring, A solvent was selected to dissolve ketoconazole, so that it could be uniformly dispersed in the system.

MECHANISM OF DRUG RELEASE:

The possible mechanism(s) through which drug releases from emulgel are:

- Diffusion through gel matrix
- Distribution of oil phase and aqueous phases
- Controlled release following Higuchi kinetics.

The gel network governs the release rate, whilst smaller droplet size can facilitate better penetration.

EVALUATION PARAMETERS:

Key evaluation tests include:

- Physical properties: Color, homogeneity, consistency
- pH: (5.5–6.5 for skin compatibility)
- Viscosity & spreadability
- Drug content uniformity
- Studies of Diffusion invitro (Franz)
- Stability studies (ICH guidelines)
- Skin irritation tests

APPLICATIONS OF EMULGELS ARE WIDELY USED IN:

- Antifungal therapy
- Anti-inflammatory drugs
- Acne and psoriasis treatment
- Local analgesic delivery

They can also deliver the drug with enough specificity to avoid systemic side effects.

RECENT ADVANCES RECENT DEVELOPMENTS INCLUDE:

- Nanoemulgels (droplets of 10–100 nm) leads to better penetration
- Smart polymers (chitosan, alginate)
- Optimization using QbD and DoE approaches

These improvements enhance potency, bioavailability and therapeutic efficacy.

COMPARISON OF TOPICAL DOSAGE FORMS

PROPERTY	EMULGEL	CREAM	OINTMENTS	HYDROGEL
Type of system	Emulsion + Gel	Emulsion (o/w or w/o)	Oil-based	Water-based
Drug type suitable	Hydrophobic drugs	Both (limited)	Mostly hydrophobic	Hydrophilic drugs
Greasiness	Non-greasy	Slightly greasy	Very greasy	Non-greasy
Spreadability	Excellent	Good	Poor	Good
Skin penetration	High	Moderate	Moderate	Low (for lipophilic drugs)
Stability	High	Moderate	High	Moderate
Drug release	Controlled (slow release)	Less controlled	Slow	Fast
Patient comfort	High	Moderate	Low	High
Wash-ability	Easy to wash	Easy	Difficult	Easy
Use in Letrozole	Suitable	Limited	Not suitable	Not suitable

MARKETED PREPARATIONS EXAMPLES INCLUDE:

- Diclofenac emulgel
- Clotrimazole emulgel
- Miconazole emulgel
- Ketoconazole formulations

These products prove to be effective and have marketability as well as Future use.

FUTURE PERSPECTIVES RESEARCH FOCUSES ON:

- Nanoemulgel systems

- Herbal and natural drug incorporation
- Improved permeation enhancers
- Cost-effective large-scale production

CONCLUSION

Emulgel is a promising topical drug delivery system, especially for hydrophobic antifungal drugs. It improves drug solubility, penetration, and patient compliance while providing controlled drug release. With advancements such as nanoemulgels and smart polymers, emulgel technology has strong potential for future pharmaceutical applications.

Topical medicine delivery will be widely employed in the upcoming years to improve patient compliance. Emulgel

is a relatively new method for topical drug delivery that works well with hydrophobic medications. Considering that it can also improve spreadability, adhesion, viscosity, and extrusion. They'll become a well-liked medicine delivery

method. Additionally, they will serve as a means of encapsulating hydrophobic medications in a gel foundation that is water soluble. A careful review of the literature has led us to the conclusion that emulgels are one of the most efficient, superior, and practical drug delivery systems since they can distribute both hydrophobic and hydrophilic medicines.

Emulgels produce greater drug release than other transdermal drug delivery systems because they are non-greasy and have a gel-like quality where the gel acts as an aqueous environment for the drug, aiding in breakdown. Emulgels are a more recent category of dosage form that are produced by trapping significant volumes of aqueous or hydroalcoholic liquid within a web of colloidal solid particles. Emulgels contain a higher aqueous component than other medications, allowing for easier drug migration through a vehicle that is virtually a liquid as well as greater drug solubility. Thus, the water phase contains the gelling ingredient that changes a conventional emulsion into an emulgel. Emulgels have gained popularity in recent years as a result of increased patient compliance. Emulgels will be a reasonable option as a topical drug administration system and a way to load hydrophobic pharmaceuticals in water soluble gel bases since they have an advantage in terms of spreadability, adhesion, viscosity, and extrusion.

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